Example 7.4

In how many ways, can the students answer 3 true or false type questions in a slip test?

Solution:

(i) Assuming that the question Q_1 is answered True, questions Q_2 and Q_3 can be answered as **TT, TF, FT** and **FF** in 4 ways.

(ii) Assuming that the question Q_1 is answered False, Q_2 and Q_3 can also be answered as **TT, TF, FT,** and **FF** in 4 ways.

Thus, as each question has only two options (True or False) in 2 ways, the number of ways of answering these 3 questions in a slip test is **8 (2 x 2 x 2) possible ways** as shown in Fig. 7.8.

Number of ways of answering the questions

	Questions		
	Q_1	Q_2	Q_3
	T/F	T/F	T/F
Answer1	T	T	T
Answer2	T	T	F
Answer3	T	F	T
Answer4	T	F	F
Answer5	F	T	T
Answer6	F	T	F
Answer7	F	F	T
Answer8	F	F	F

$[(1\ (Q_1) \times 2\ (T/F)) \times (1\ (Q_2) \times 2\ (T/F)) \times (1\ (Q_3) \times 2\ (T/F))] = 8$ ways

Fig. 7.8

Example 7.5

Madhan wants to a buy a new car. The following choices are available for him.

- There are **2 types** of cars as shown in the Fig. 7.9
- There are **5 colours** available in each type as shown in Fig. 7.9.
- There are **3 models** available in each colour
  - (i) **GL** (standard model)
  - (ii) **SS** (sports model)
  - (iii) **SL** (luxury model)

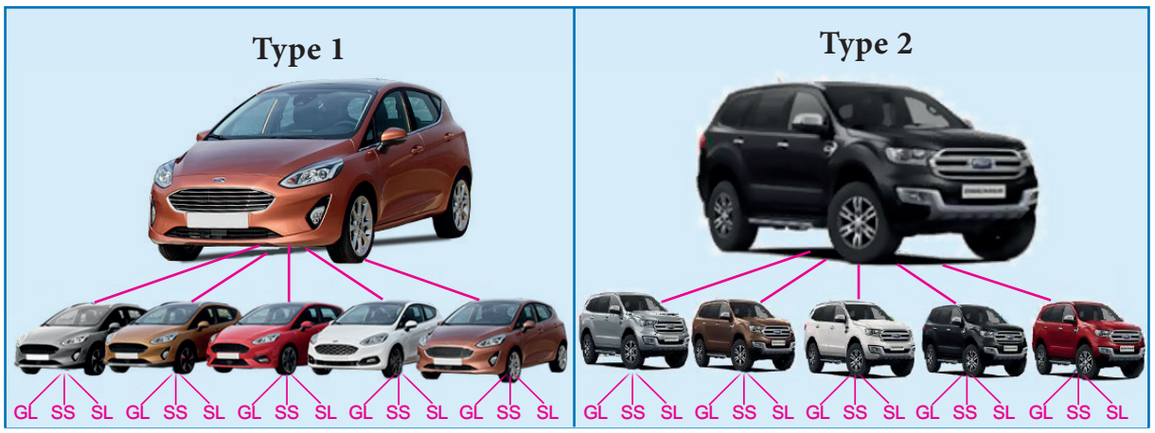


Fig. 7.9

(i) In how many different ways can Madhan buy any one of the new car?

(ii) If the **white colour** is not available in **Type 2**, then in how many ways can Madan buy a new car among the given option?

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